AIS – AURELIA INDUSTRIAL SYSTEMS

AIS-200 USER MANUAL

Aurelia Industrial Systems (AIS)

AIS-200 User Manual
Industrial Assembly System
Document ID: AIS-MAN-200
Version: 2.1
Effective Date: January 15, 2026
Category: User Manual
Language: English
Classification: Internal / Customer Reference

---

1. Introduction

The AIS-200 is an industrial automated assembly system designed for medium- and high-volume manufacturing environments.

The machine combines robotic automation, an advanced Human-Machine Interface, remote diagnostic capabilities, and redundant safety systems.

The standard AIS-200 configuration provides a nominal maximum production capacity of:

80 units per minute.

Actual performance depends on the production configuration, product characteristics, tooling, material flow, and connected equipment.

---

2. Intended Use

The AIS-200 is intended for automated industrial assembly applications.

Typical applications include:

- Component assembly.
- Robotic part handling.
- Automated positioning.

- High-volume production.
- Integrated manufacturing cells.

The machine must only be used within its documented technical and environmental limits.

---

3. Safety Requirements

Only trained and authorized personnel may operate the AIS-200.

Before operation, verify:

- Emergency-stop devices.
- Safety sensors.
- Guard interlocks.
- Safety-controller status.
- Machine operating area.
- Robot safety zones.

Operators must never bypass or modify safety functions.

---

4. Redundant Safety System

The AIS-200 includes redundant safety controls designed to monitor critical safety conditions.

Safety-related functions include:

- Emergency stop.
- Guard monitoring.
- Safety sensors.
- Controlled machine stopping.
- Safety-controller diagnostics.

If a safety function reports an abnormal condition, production must not continue until the condition has been correctly evaluated.

---

5. Installation

Installation and commissioning must be performed by AIS personnel or authorized technicians.

Verify:

- Machine positioning.
- Electrical connections.
- Protective earth.
- Safety systems.
- Network connectivity.
- Cooling-system connections.
- Environmental conditions.
- Robot operating area.

The installation must comply with the approved project documentation.

---

6. Electrical Requirements

The standard AIS-200 electrical configuration is:

400 VAC, three-phase, 50/60 Hz.

Electrical work must be performed by qualified personnel.

---

7. Startup Procedure

Before startup:

1. Verify that the production area is clear.
2. Check safety guards.
3. Verify emergency-stop status.
4. Verify safety-controller status.
5. Switch on the main power supply.
6. Allow the control system to initialize.

7. Check the HMI for active alarms.
8. Verify cooling-system status.
9. Complete the machine homing procedure.
10. Enable automatic production.

If an abnormal alarm occurs, do not repeatedly restart the machine.

---

8. Human-Machine Interface

The AIS-200 HMI provides access to:

- Machine status.
- Production status.
- Alarm history.
- Error codes.
- Maintenance counters.
- Diagnostic information.
- Selected configuration parameters.

Only authorized users may modify configuration parameters.

---

9. Normal Operation

The nominal maximum production capacity of the AIS-200 is:

80 units/minute.

Operators should monitor:

- Machine status.
- Production rate.
- Robot status.
- Safety status.
- Cooling-system status.

- Active alarms.
- Product quality.

Unexpected temperature, vibration, noise, or repeated alarms should be investigated.

---

10. Shutdown Procedure

For normal shutdown:

1. Stop automatic production.
2. Allow the machine to complete the current safe cycle.
3. Disable automatic operation.
4. Confirm safe machine status.
5. Perform the required end-of-shift inspection.
6. Follow the normal shutdown procedure.

For maintenance, apply the approved energy-isolation procedure.

---

11. Preventive Maintenance

The **current approved preventive-maintenance interval for the AIS-200 is 2,000 operating hours**.

Maintenance should include:

- Safety-system inspection.
- Robot inspection.
- Mechanical inspection.
- Sensor inspection.
- Cooling-system inspection.
- Electrical-cabinet inspection.
- Lubrication where applicable.
- Alarm-history review.

Maintenance records must be stored against the machine serial number.

12. Cooling System

The AIS-200 contains a cooling system used to maintain operating temperatures within the documented range.

Operators should periodically verify:

- Cooling-system status.
- Coolant level.
- Visible coolant flow conditions.
- Abnormal temperature indications.
- Leakage or visible damage.

The cooling system must not be bypassed.

---

13. Error Code E-204

Description

E-204 indicates that the cooling-system temperature is above the configured threshold.

The alarm may occur during startup or operation when the monitored cooling-system temperature exceeds the configured limit.

Required Action

When E-204 is displayed:

1. Stop the machine safely.
2. Do not continue production while the alarm remains active.
3. Check the cooling-system status.
4. Verify the coolant level.
5. Check for obvious leakage or abnormal conditions.
6. Verify that the cooling unit is operating normally.
7. Follow the approved recovery procedure.

If the alarm persists after the approved checks, contact AIS Technical Support.

---

14. E-204 Safety Warning

Do not:

- Bypass the temperature protection.
- Disable the alarm.
- Modify the temperature threshold without authorization.
- Continue production solely to avoid downtime.
- Open equipment requiring qualified technical intervention.

If the machine shows signs of overheating, smoke, leakage, or another abnormal condition, stop the machine and follow the applicable safety procedure.

---

15. Technical Support Information for E-204

When contacting Technical Support about E-204, provide:

- Machine serial number.
- Machine operating hours.
- Current machine state.
- Error occurrence time.
- Cooling-system status.
- Coolant level.
- Previous occurrences of E-204.
- Production impact.

AIS Technical Support may request additional diagnostic information.

---

16. General Troubleshooting

Machine Does Not Start

Check:

- Emergency-stop status.

- Safety guards.
- Safety-controller status.
- Active HMI alarms.
- Cooling-system readiness.
- Initialization status.

Machine Stops During Production

Record:

- Error code.
- Machine state.
- Production impact.
- Recent operator actions.

Follow the applicable troubleshooting procedure before restarting the machine.

---

17. Technical Specifications

Parameter	Standard Value
Product	AIS-200
Machine type	Industrial Assembly System
Nominal capacity	80 units/min
Power supply	400 VAC, three-phase, 50/60 Hz
Operating temperature	5–35 °C
PLC	AIS Control PLC-200
Communication	Industrial Ethernet, OPC UA, digital I/O
Safety	Redundant safety controller
Preventive maintenance	Every 2,000 operating hours

18. Maintenance Safety

Maintenance must be performed by trained and authorized personnel.

Before maintenance:

1. Stop production.
2. Place the machine in a safe state.
3. Isolate energy sources.
4. Verify the absence of hazardous energy.
5. Follow the approved maintenance procedure.

Cooling-system intervention may require qualified technical personnel.

---

19. Remote Diagnostics

Authorized AIS Technical Support personnel may use remote diagnostics when supported and contractually permitted.

Remote diagnostics may include:

- Alarm history.
- Controller status.
- Selected temperature parameters.
- Communication status.
- Diagnostic information.

Remote access is subject to AIS authentication, authorization, and security requirements.

---

20. Technical Support

Contact Technical Support when:

- E-204 persists after approved checks.
- The machine repeatedly reports the same alarm.
- A safety-related condition occurs.

- Operator-level troubleshooting does not resolve the issue.
- Qualified technical intervention is required.

---

21. Document Control

Version	Effective Date	Description
2.0	2025-01-10	Updated machine operation procedures
2.1	2026-01-15	Updated cooling-system troubleshooting and maintenance information

Current approved version: 2.1